

A Packing of 9 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 9 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.699735. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.666374, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 9 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.699735 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 9$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.666374, obtained by deleting 1 of the 10 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 9/8 = 2.700000$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N9.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
269973581349864111622613701047015553886662896827708881458765078122689496
560713356523732656510922067195947646942385837418515808455109543707260869
6269977569964205034516380712767893598981323625993
```

denominator:

```
100000000000304433862397466704639467934267741163143547784592020630025676
492321297848589861770737049945944230617316011010350011444521673018036452
2679655040170045033219177326422395191382067089434
```

B Coordinates

9 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N9.txt`.

```
-38603676730 166236957383 -146752670489 -167393496380 074500633 95549120180 -743533484813 112691604432 72430150197 -162403698500 -103531418804 -58222816780
-290352105256 -448513297361 243350615173 -597372981989 3819398 35477844824 -345450536769 -551402375588 127913854927 309865449936 -675937393370 -21553066043
473940840361 11636280492 121074757238 -35278720200 500032841131 381230809096 420638275688 -125597628286 14684565468 202097362533 75381427306 18558891565
300429192725 -741165897290 -413483449840 146195011763 384343501 236886288568 -111827267442 -42233638993 583380833001 118822752904 138619350006 469513684578
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-342689464970 252549392017 -33199424593 -179860104200 577709762 530911161535 32319111118 63858705090 298175272394 -218622705691 144213498790 248697244620
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```